

MACIT2506212882025Mini8

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1 Miniproject Assessed Exercise

2 Level 8

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2.5 Version 1

2.5.1 Double or Quit Quiz program

Write a procedural program game to play where players have to answer questions with short answers (NOT multiple choice). The person starts with an amount of money in a pot. With each correct answer the person doubles their winnings. If they get an answer wrong then they halve their winnings

2.6 Justification that the program passes this level

This is based on tick boxes at start of the question description for miniproject program level 1 - I can EXPLAIN HOW MY PROGRAM WORKS - I wrote this program myself - My program compiles and runs in JHUB - My program screen output and keyboard input - My program includes methods doing a well-defined task and uses a sequence of method calls - My program include simple comments including my name, student number and date - I have written a literate version of the code - documenting each method - including a test plan for each method - included a test plan for the whole program - included a full version of the program at the end - My program defines all variables inside methods - My program includes variables, assignment and expressions - My program defines and uses at least one method that returns a result - My program attempts indentation

2.6.1 print

What it does Simplify's `System.out.println()` to `print()`

Implementation (how it works) “ - `{System.out.println()}` displays a String on the screen

```
[6]: // Print a message to the user  
    //
```

```
public static void print(String message) {
    System.out.println(message);

    return;
} // END print
```

Testing

```
[2]: print("What's your name, and how old are you?")
```

What's your name, and how old are you?

2.6.2 inputString

What it does Print's a question, and returns the answer given by the user to store into a variable as long as it's a String.

Implementation (how it works) ““ - {Scanner scanner = new Scanner(System.in)} allows keyboard input from the user - {System.out.println()} displays a String on the screen - {scanner.nextLine()} reads the user's input

```
[7]: // Ask's a question which requires a String input, and returns the answer the
    ↪user inputs
    //
    public static String inputString(String message) {
        Scanner scanner = new Scanner(System.in);
        System.out.println(message);

        return scanner.nextLine();
    } // END inputString
```

Testing

```
[6]: inputString("What's your name?")
```

What's your name?

Suleyman

```
[6]: Suleyman
```

2.6.3 startNewOrLoad

```
[8]: // Ask's user if they wish to start a new quiz or load the save file of a quiz
    ↪which has already started
    //
    public static String startNewOrLoad () {
```

```

        return inputString("Type NEW if you want to start a new quiz or LOAD if you
        ↪already have a saved file.");
    } // END startNewOrLoad

```

2.6.4 theGreetings

What it does

- Quiz host introduces itself.
- States that if the user wants to start, they must agree to all the rules.

Implementation (how it works)

Uses {print} to give a series of print statements.

```

[9]: // Print a quizBot greeting message
    //
    public static void theGreetings()
    {
        print("Greetings. I am Dr Quinn, your host for today.");
        print("I'll be with you throughout the whole Quiz, so it's best you grow a
        ↪likin' for me.");
        print("Before we begin, you all need to review the Terms & Conditions
        ↪below.");

        return;
    } // END theGreetings

```

Testing

```

[9]: // TEST CODE for method 1 [theGreetings]
    theGreetings();

```

Greetings. I am Dr Quinn, your host for today.

I'll be with you throughout the whole Quiz, so it's best you grow a likin' for me.

Before we begin, you all need to review the Terms & Conditions below.

2.6.5 rulesAndAcceptance

What it does

- Gives a list of the Terms & Conditions.
- By giving Dr Quinn your name, user agrees to all rules and wishes to continue to the Quiz.

Implementation (how it works)

- Uses {print} to give a series of print statements.
- Uses {Scanner scanner = new Scanner(System.in)} to be able to take in keyboard input from user.
- Uses {String =} to declare string variables.

- Uses {scanner.nextLine()} and waits for user to input something, which is then assigned to a

```
[10]: // Tells the user the Rules and by inputting anything, they agree to them
      ↪completely.
      //
      public static void rulesAndAcceptance()
      {
          String accept;
          Scanner scanner = new Scanner(System.in);

          print(" ");
          print("Here are the T&C's for today's quiz.");
          print(" ");

          print("1. Each player begins with £2000 in the pot.");
          print("2. A correct answer DOUBLES the pot. An incorrect answer HALVES the
      ↪pot.");
          print("3. Each player is given 2 attempts per question.");
          print("4. The game continues until all the questions are finished.");
          print("5. Answers MUST be entered clearly and in CAPITALS. Spelling
      ↪mistakes WILL be judged incorrect, and cannot be compensated for.");
          print("6. Cheating is strictly FORBIDDEN. This includes working together,
      ↪searching things up online or receiving help from others. Any form of it and
      ↪you're outta here.");
          print("7. Last but not least: Have fun.");

          print(" ");
          print("[To agree and continue, type anything.]");
          print(" ");

          accept = scanner.nextLine();

          print("Great, let's get this quiz underway, shall we?");
          print(" ");

          return;
      } // END rulesAndAcceptance
```

```
[11]: // TEST CODE for method 2 [rulesAndName]
      rulesAndAcceptance();
```

Here are the T&C's for today's quiz.

1. Each player begins with £2000 in the pot.
2. A correct answer DOUBLES the pot. An incorrect answer HALVES the pot.
3. Each player is given 2 attempts per question.

4. The game continues until all the questions are finished.
5. Answers MUST be entered clearly and in CAPITALS. Spelling mistakes WILL be judged incorrect, and cannot be compensated for.
6. Cheating is strictly FORBIDDEN. This includes working together, searching things up online or receiving help from others. Any form of it and you're outta here.
7. Last but not least: Have fun.

[To agree and continue, type anything.]

x

Great, let's get this quiz underway, shall we?

2.6.6 double_pot

What it does It receives the value of the current pot and doubles it, returning that value.

Implementation (how it works)

- (int pot_to_double) means the method has to be called with an integer to actually work [e.g. double_pot(2000)]
- [return (pot_to_double * 2)] multiplies the integer the method has been called with by 2, and returns the result

```
[11]: // Receives the value of the current pot and doubles it.
//
public static int double_pot (int pot_to_double)
{
    return (pot_to_double * 2);
} // END double_pot
```

Testing

```
[13]: // TEST CODE for method 2.5.1 [double_pot]

double_pot(2000);
```

[13]: 4000

2.6.7 halve_pot

What it does It receives the value of the current pot and halves it, returning that value.

Implementation (how it works)

- (int pot_to_halve) means the method has to be called with an integer to actually work [e.g. halve_pot(2000)]
- [return (pot_to_halve / 2)] divides the integer the method has been called with by 2, and returns the result

```
[12]: // Recieves the value of the current pot and halves it.
//
public static int halve_pot (int pot_to_halve)
{
    return (pot_to_halve / 2);
} // END halve_pot
```

Testing

```
[15]: // TEST CODE for method 2.5.2 [halve_pot]

halve_pot(2000);
```

[15]: 1000

2.6.8 displayPots

What it does Prints to the screen the amount each player currently has in their pot

Implementation (how it works) ““ - Uses {print} to give a series of print statements.

```
[13]: // Recieves the array of pots and prints them out
//
public static void displayPots(int[] pots) {

    print("Currently: ");
    print("Player 1 has : £" + pots[0]);
    print("Player 2 has : £" + pots[1]);
    print("Player 3 has : £" + pots[2]);
    print("Player 4 has : £" + pots[3]);
    print("In their pots.");

    return;
} // END displayPots
```

Testing

```
[14]: int [] pots = {2000, 2000, 2000, 2000};

displayPots(pots);
```

Currently:
 Player 1 has : £2000
 Player 2 has : £2000
 Player 3 has : £2000
 Player 4 has : £2000
 In their pots.

2.6.9 Question

What it does Creates a record which holds the question and its answer

Implementation (how it works) ““ - Uses {class _____} to create the record - Uses {variable type _____} to create the fields

```
[18]: // Creates the record Question which stores the Questions and their Answers
//
class Question {

    String questionText;
    String answer;

} // END Question
```

2.6.10 createQuestion

What it does Creates a variable of data type Question

Implementation (how it works) ““ - {Question x = new Question()} is needed to create a new record - {x.field = _____} to assign the arguments into the fields

```
[19]: // Creates a new Question record, storing the Question and Answer with the
      ↪arguments given
//
public static Question createQuestion(String q, String a) {

    Question x = new Question();

    x.questionText = q;
    x.answer = a;

    return x;
} // END createQuestion
```

Testing

```
[23]: Question [] EXAMPLEQuestionList = new Question [1];

EXAMPLEQuestionList [0] = createQuestion("What's my name?", "Suleyman");
```

```
[23]: REPL.$JShell$28$Question@3826777c
```

2.6.11 getQuestionText

What it does Returns the question text stored in a Question record

Implementation (how it works) ““ - Uses the parameter {Question x} so the method requires a Question record to be chosen - {x.questionText} is returned, meaning whatever is stored in the questionText field of the Question record which was given as an argument

```
[25]: // Returns whatever is stored in the questionText field of a Question record  
//  
public static String getQuestionText (Question x) {  
  
    return x.questionText;  
} // END getQuestionText
```

Testing

```
[26]: String questionEXAMPLE;  
  
questionEXAMPLE = getQuestionText(EXAMPLEquestionList[0]);  
  
print(questionEXAMPLE);
```

What's my name?

2.6.12 getAnswer

What it does Returns the answer stored in a Question record

Implementation (how it works) ““ - Uses the parameter {Question x} so the method requires a Question record to be chosen - {x.answer} is returned, meaning whatever is stored in the questionText field of the Question record which was given as an argument

```
[28]: // Returns whatever is stored in the answer field of a Question record  
//  
public static String getAnswer (Question x) {  
  
    return x.answer;  
} // END getAnswer
```

Testing

```
[29]: String answerEXAMPLE;  
  
answerEXAMPLE = getAnswer(EXAMPLEquestionList[0]);  
  
print(answerEXAMPLE);
```

Suleyman

2.6.13 saveGameState

What it does Saves the current question number and the pot values for each player

Implementation (how it works) ““ - {PrintWriter game_file = new PrintWriter(new FileWriter(filename))} creates a file named savedFile - {game_file.println(Qnumb)} saves the value in Qnumb as the first line of the file - {for} loop used to go through the current pots and save them into the next lines of the file

```
[31]: // Saves the current current Pot values and the Question number into the File
      ↪ savedFile.txt
      //
      public static void saveGameState (int Qnumb, int [] pots) throws IOException {

          String filename = "savedFile.txt";

          PrintWriter game_file = new PrintWriter(new FileWriter(filename));

          game_file.println(Qnumb);

          for (int i = 0; i < 4; i++) {

              game_file.println(pots[i]);

          }

          game_file.close();

          return;
      } // END saveGameState
```

Testing

```
[32]: saveGameState(0, pots);
```

2.6.14 loadGameStateQnumb

What it does Loads the question number from the saved game

Implementation (how it works) ““ - {BufferedReader game_file = new BufferedReader(new FileReader(filename))} reads the file named savedFile - {Qnumb = Integer.parseInt(game_file.readLine())} converts the number in Qnumb into an integer and returns it

```
[34]: // Accesses the Question number value stored in the File savedFile.txt
      //
      public static int loadGameStateQnumb () throws IOException {

          String filename = "savedFile.txt";
          int Qnumb = 0;

          BufferedReader game_file = new BufferedReader(new FileReader(filename));
```

```

    Qnumb = Integer.parseInt(game_file.readLine());

    game_file.close();

    return Qnumb;
} // END loadGameState

```

Testing

```

[37]: int Qnumb;

Qnumb = loadGameStateQnumb();

System.out.println(Qnumb);

```

0

2.6.15 loadGameStatePots

What it does Loads the current pots from the saved game

Implementation (how it works) ““ - {BufferedReader game_file = new BufferedReader(new FileReader(filename))} reads the file named savedFile - {for} loop is used to go through each pot value, converting them into integers in order to return them

```

[38]: // Accesses the Pot values stored in the File savedFile.txt
//
public static int [] loadGameStatePots (int [] pots) throws IOException {

    String filename = "savedFile.txt";
    int Qnumb = 0;

    BufferedReader game_file = new BufferedReader(new FileReader(filename));

    Qnumb = Integer.parseInt(game_file.readLine());

    for (int i = 0; i < 4; i++) {

        pots[i] = Integer.parseInt(game_file.readLine());
    }

    game_file.close();

    return pots;
} // END loadGameState

```

Testing

```
[40]: pots = loadGameStatePots(pots);

displayPots(pots);
```

Currently:

```
Player 1 has : £2000
Player 2 has : £2000
Player 3 has : £2000
Player 4 has : £2000
In their pots.
```

2.6.16 runQuiz

What it does Asks the questions and gives each player 2 attempts to answer the question correctly.

Implementation (how it works) “ - Uses parameters {String [] questions, String [] answers, int[] pots} so the method can work with arrays passed from other methods - {print} prints a message to the screen - {for} loop to move through each quiz question in turn. - Nested {for} loop so each of the 4 players gets a turn to answer every question. - {while} loop inside a {for} loop to allow each player multiple [2] attempts for each question

```
[45]: // Recieves all the arrays as arguments, then asks the quiz questions in order
      ↪ - changing each players pot depending if they get their attempts right or
      ↪ wrong
      //
public static void runQuiz(Question [] questionList, int[] pots, int Qnumb)
    ↪ throws IOException {

    for (int i = Qnumb; i<=4; i++) {

        saveGameState(Qnumb, pots);
        displayPots(pots);

        print("Question " + (Qnumb) + ": " + getQuestionText(questionList[Qnumb
    ↪ - 1]));

        for (int x = 0; x<4; x++) {

            print("Player " + (x+1) + "'s go: ");

            String answer;
            int attempt = 1;
            boolean correct = false;
            final int NUMBER_OF_ATTEMPTS = 2;

            while (attempt <= NUMBER_OF_ATTEMPTS && correct == false) {
```

```

        answer = inputString("What is your attempt number: " + attempt + "?");

        if (answer.equals(getAnswer(questionList[Qnumb-1]))) {

            print("That is the right answer.");
            pots[x] = double_pot(pots[x]);
            correct = true;
        }
        else {

            print("That is the wrong answer.");
            pots[x] = halve_pot(pots[x]);
            attempt = attempt + 1;
        }
    }
    Qnumb = Qnumb + 1;
}
} // END runQuiz

```

2.6.17 quizBotLvl6

What it does Calls all the methods in order, creates the arrays needed and prints the final earnings for each player

Implementation (how it works) ““ - {print} prints a message to the screen - {_____} [] = { , , } creates an array of type _____, and of length __ depending on how many variables you place into the array

```

[25]: // A quizBot - Prints a welcome message, the rules, asks for the user's name,
      ↪states the initial pot value, gives the quiz questions, halves the pot if
      ↪the answers inputted are wrong
      // and doubles the pot if the answers inputted are right - and then prints the
      ↪final winnings.
public static void quizBotLvl6 () throws IOException
{
    String option = "";
    option = startNewOrLoad();

    int Qnumb = 1;

    int [] pots = {2000, 2000, 2000, 2000};

    if (option.equals("NEW")) {

```

```

    theGreetings();
    rulesAndAcceptance();
}
else if (option.equals("LOAD")) {
    Qnumb = loadGameStateQnumb();
    pots = loadGameStatePots(pots);
}
else {
    print("ONLY INPUT 'NEW' OR 'LOAD'. AND MAKE SURE IT'S IN CAPITALS");

    option = startNewOrLoad();
}

Question [] questionList = new Question [4];

questionList [0] = createQuestion("[Q1] What is the chemical symbol for_
↪water? [ALL CAPITALS]", "H2O");
questionList [1] = createQuestion("[Q2] Fill in the blank: The _ _ _ _ _ _ _ _
↪Ocean is the largest ocean on earth. [ALL CAPITALS]", "PACIFIC");
questionList [2] = createQuestion("[Q3] What country gifted the famous_
↪'Statue of Liberty' to the USA? [ALL CAPITALS]", "FRANCE");
questionList [3] = createQuestion("[Q4] What is the only mammal capable of_
↪true flight? [ALL CAPITALS] | [PLURAL ANSWER WANTED]", "BATS");

runQuiz(questionList, pots, Qnumb);

print("Player 1 has earned: £" + pots[0]);
print("Player 2 has earned: £" + pots[1]);
print("Player 3 has earned: £" + pots[2]);
print("Player 4 has earned: £" + pots[3]);
print("Well done to all!");

} // END quizBotLvl6

```

Testing

[26]: `quizBotLvl6();`

Type NEW if you want to start a new quiz or LOAD if you already have a saved file.

NEW

Greetings. I am Dr Quinn, your host for today.

I'll be with you throughout the whole Quiz, so it's best you grow a likin' for me.

Before we begin, you all need to review the Terms & Conditions below.

Here are the T&C's for today's quiz.

1. Each player begins with £2000 in the pot.
2. A correct answer DOUBLES the pot. An incorrect answer HALVES the pot.
3. Each player is given 2 attempts per question.
4. The game continues until all the questions are finished.
5. Answers MUST be entered clearly and in CAPITALS. Spelling mistakes WILL be judged incorrect, and cannot be compensated for.
6. Cheating is strictly FORBIDDEN. This includes working together, searching things up online or receiving help from others. Any form of it and you're outta here.
7. Last but not least: Have fun.

[To agree and continue, type anything.]

Cool

Great, let's get this quiz underway, shall we?

Currently:

Player 1 has : £2000

Player 2 has : £2000

Player 3 has : £2000

Player 4 has : £2000

In their pots.

Question 1: [Q1] What is the chemical symbol for water? [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 2's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 3's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 4's go:

What is your attempt number: 1?

H2O

That is the right answer.

Currently:

Player 1 has : £4000

Player 2 has : £4000

Player 3 has : £4000

Player 4 has : £4000

In their pots.

Question 2: [Q2] Fill in the blank: The _____ Ocean is the largest ocean on earth. [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 2's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 3's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 4's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Currently:

Player 1 has : £8000

Player 2 has : £8000

Player 3 has : £8000

Player 4 has : £8000

In their pots.

Question 3: [Q3] What country gifted the famous 'Statue of Liberty' to the USA? [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 2's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 3's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 4's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Currently:

Player 1 has : £16000

Player 2 has : £16000

Player 3 has : £16000

Player 4 has : £16000

In their pots.

Question 4: [Q4] What is the only mammal capable of true flight? [ALL CAPITALS]
| [PLURAL ANSWER WANTED]

Player 1's go:

What is your attempt number: 1?

BATS

That is the right answer.

Player 2's go:

What is your attempt number: 1?

BATS

That is the right answer.

Player 3's go:

What is your attempt number: 1?

BATS

That is the right answer.

Player 4's go:

What is your attempt number: 1?

BATS

That is the right answer.

Player 1 has earned: £32000

Player 2 has earned: £32000

Player 3 has earned: £32000

Player 4 has earned: £32000

Well done to all!

3 The complete program

“ // Cut and paste this version as a Java file and compile and run it locally to check it works.
// The full program should include // - libraries (where needed), // - main method and // - class
definition // - Each method should have a comment explaining its purpose // - The program should
start with comments as below


```

[3]: /* *****
    @author    Suleyman Macit
    @SID       250621288
    @date      30 November 2025
    @version   1

    Miniproject Level 6
    A quizBot program in the style of 'Double or Quit'. It gives the users the
    ↪rules and 4 questions. Question answers are checked with if statements and
    ↪each
    question has 2 attempts using while loops.
    *****/

import java.util.Scanner; // Needed to make Scanner available
import java.io.*;

// Creates the record Question which stores the Questions and their Answers
//
class Question {

    String questionText;
    String answer;

} // END Question

public class quiz6
{
    public static void main (String [] a) throws IOException
    {
        quizBotLvl6 ();
        return;
    } // END main

    // Print a message to the user
    //
    public static void print(String message) {
        System.out.println(message);

        return;
    } // END print

    // Ask's a question which requires a String input, and returns the
    ↪answer the user inputs
    //
    public static String inputString(String message) {
        Scanner scanner = new Scanner(System.in);
        System.out.println(message);
    }

```

```

        return scanner.nextLine();
    } // END inputString

    // Print a quizBot greeting message
    //
    public static void theGreetings()
    {
        print("Greetings. I am Dr Quinn, your host for today.");
        print("I'll be with you throughout the whole Quiz, so it's best you
        grow a likin' for me.");
        print("Before we begin, you all need to review the Terms & Conditions
        below.");

        return;
    } // END theGreetings

    // Tells the user the Rules and by inputting anything, they agree to them
    completely.
    //
    public static void rulesAndAcceptance()
    {
        String accept;
        Scanner scanner = new Scanner(System.in);

        print(" ");
        print("Here are the T&C's for today's quiz.");
        print(" ");

        print("1. Each player begins with £2000 in the pot.");
        print("2. A correct answer DOUBLES the pot. An incorrect answer HALVES
        the pot.");
        print("3. Each player is given 2 attempts per question.");
        print("4. The game continues until all the questions are finished.");
        print("5. Answers MUST be entered clearly and in CAPITALS. Spelling
        mistakes WILL be judged incorrect, and cannot be compensated for.");
        print("6. Cheating is strictly FORBIDDEN. This includes working
        together, searching things up online or receiving help from others. Any form
        of it and you're outta here.");
        print("7. Last but not least: Have fun.");

        print(" ");
        print("[To agree and continue, type anything.]");
        print(" ");

        accept = scanner.nextLine();
    }

```

```

    print("Great, let's get this quiz underway, shall we?");
    print(" ");

    return;
} // END rulesAndAcceptance

    // Recieves the value of the current pot and doubles it.
    //
    public static int double_pot (int pot_to_double)
    {
        return (pot_to_double * 2);
    } // END double_pot

    // Recieves the value of the current pot and halves it.
    //
    public static int halve_pot (int pot_to_halve)
    {
        return (pot_to_halve / 2);
    } // END halve_pot

    // Recieves the array of pots and prints them out
    //
    public static void displayPots(int[] pots) {

        print("Currently: ");
        print("Player 1 has : £" + pots[0]);
        print("Player 2 has : £" + pots[1]);
        print("Player 3 has : £" + pots[2]);
        print("Player 4 has : £" + pots[3]);
        print("In their pots.");

        return;
    } // END displayPots

    // Creates a new Question record, storing the Question and Answer with
    → the arguments given
    //
    public static Question createQuestion(String q, String a) {

        Question x = new Question();

        x.questionText = q;
        x.answer = a;
    }

```

```

        return x;
    } // END createQuestion

    // Returns whatever is stored in the questionText field of a Question
    ↪record
    //
    public static String getQuestionText (Question x) {

        return x.questionText;
    } // END getQuestionText

    // Returns whatever is stored in the answer field of a Question record
    //
    public static String getAnswer (Question x) {

        return x.answer;
    } // END getAnswer

    // Ask's user if they wish to start a new quiz or load the save file of
    ↪a quiz which has already started
    //
    public static String startNewOrLoad () {

        return inputString("Type NEW if you want to start a new quiz or
    ↪LOAD if you already have a saved file.");
    } // END startNewOrLoad

    // Saves the current current Pot values and the Question number
    ↪into the File savedFile.txt
    //
    public static void saveGameState (int Qnumb, int [] pots) throws
    ↪IOException {

        String filename = "savedFile.txt";

        PrintWriter game_file = new PrintWriter(new
    ↪FileWriter(filename));

        game_file.println(Qnumb);

        for (int i = 0; i < 4; i++) {

            game_file.println(pots[i]);
        }

        game_file.close();
    }

```

```

        return;
    } // END saveGameState

    // Accesses the Question numb values stored in the File savedFile.
↪txt
    //
    public static int loadGameStateQnumb () throws IOException {

        String filename = "savedFile.txt";
        int Qnumb = 0;

        BufferedReader game_file = new BufferedReader(new
↪FileReader(filename));

        Qnumb = Integer.parseInt(game_file.readLine());

        game_file.close();

        return Qnumb;
    } // END loadGameState

    // Accesses the Pot values stored in the File savedFile.txt
    //
    public static int [] loadGameStatePots (int [] pots) throws
↪IOException {

        String filename = "savedFile.txt";
        int Qnumb = 0;

        BufferedReader game_file = new BufferedReader(new
↪FileReader(filename));

        Qnumb = Integer.parseInt(game_file.readLine());

        for (int i = 0; i < 4; i++) {

            pots[i] = Integer.parseInt(game_file.readLine());
        }

        game_file.close();

        return pots;
    } // END loadGameState

```

```

    // Recieves all the arrays as arguments, then asks the quiz questions in
    ↪order - changing each players pot depending if they get their attempts right
    ↪or wrong
    //
    public static void runQuiz(Question [] questionList, int[] pots, int Qnumb)
    ↪throws IOException {

        for (int i = Qnumb; i<=4; i++) {

            saveGameState(Qnumb, pots);
            displayPots(pots);

            print("Question " + (Qnumb) + ": " +
    ↪getText(questionList[Qnumb - 1]));

            for (int x = 0; x<4; x++) {

                print("Player " + (x+1) + "'s go: ");

                String answer;
                int attempt = 1;
                boolean correct = false;
                final int NUMBER_OF_ATTEMPTS = 2;

                while (attempt <= NUMBER_OF_ATTEMPTS && correct == false) {

                    answer = inputString("What is your attempt number: " +
    ↪attempt + "?");

                    if (answer.equals(getAnswer(questionList[Qnumb-1]))) {

                        print("That is the right answer.");
                        pots[x] = double_pot(pots[x]);
                        correct = true;
                    }
                    else {

                        print("That is the wrong answer.");
                        pots[x] = halve_pot(pots[x]);
                        attempt = attempt + 1;
                    }
                }
                Qnumb = Qnumb + 1;
            }
        } // END runQuiz

```

```

    // A quizBot - Prints a welcome message, the rules, asks for the user's
    ↪name, states the initial pot value, gives the quiz questions, halves the pot,
    ↪if the answers inputted are wrong
    // and doubles the pot if the answers inputted are right - and then prints
    ↪the final winnings.
    public static void quizBotLvl6 () throws IOException
    {
        String option = "";
        option = startNewOrLoad();

        int Qnumb = 1;

        int [] pots = {2000, 2000, 2000, 2000};

        if (option.equals("NEW")) {

            theGreetings();
            rulesAndAcceptance();
        }
        else if (option.equals("LOAD")) {
            Qnumb = loadGameStateQnumb();
            pots = loadGameStatePots(pots);
        }
        else {
            print("ONLY INPUT 'NEW' OR 'LOAD'. AND MAKE SURE IT'S IN CAPITALS");

            option = startNewOrLoad();
        }

        Question [] questionList = new Question [4];

        questionList [0] = createQuestion("[Q1] What is the chemical symbol for
        ↪water? [ALL CAPITALS]", "H2O");
        questionList [1] = createQuestion("[Q2] Fill in the blank: The _____
        ↪Ocean is the largest ocean on earth. [ALL CAPITALS]", "PACIFIC");
        questionList [2] = createQuestion("[Q3] What country gifted the famous
        ↪'Statue of Liberty' to the USA? [ALL CAPITALS]", "FRANCE");
        questionList [3] = createQuestion("[Q4] What is the only mammal capable
        ↪of true flight? [ALL CAPITALS] | [PLURAL ANSWER WANTED]", "BATS");

        runQuiz(questionList, pots, Qnumb);

        print("Player 1 has earned: £" + pots[0]);
        print("Player 2 has earned: £" + pots[1]);
        print("Player 3 has earned: £" + pots[2]);
    }

```

```

        print("Player 4 has earned: £" + pots[3]);
        print("Well done to all!");

    } // END quizBotLvl6
} // END class

```

3.1 Testing the complete program

Run the following test calls for each test in the test plan to run the complete program.

3.1.1 Test Plan: Test 1

If the following input is given by the user when asked in turn:

```

NEW |   Cool |   | Player 1: H2O, PACIFIC, FRANCE, BATS   | Player 2: H2O, PACIFIC,
FRANCE, BATS   | Player 3: H2O, PACIFIC, FRANCE, BATS   | Player 4: H2O, PACIFIC,
FRANCE, BATS ...

```

```
[33]: quiz6.main(null);
```

Type NEW if you want to start a new quiz or LOAD if you already have a saved file.

NEW

Greetings. I am Dr Quinn, your host for today.

I'll be with you throughout the whole Quiz, so it's best you grow a likin' for me.

Before we begin, you all need to review the Terms & Conditions below.

Here are the T&C's for today's quiz.

1. Each player begins with £2000 in the pot.
2. A correct answer DOUBLES the pot. An incorrect answer HALVES the pot.
3. Each player is given 2 attempts per question.
4. The game continues until all the questions are finished.
5. Answers MUST be entered clearly and in CAPITALS. Spelling mistakes WILL be judged incorrect, and cannot be compensated for.
6. Cheating is strictly FORBIDDEN. This includes working together, searching things up online or receiving help from others. Any form of it and you're outta here.
7. Last but not least: Have fun.

[To agree and continue, type anything.]

Cool

Great, let's get this quiz underway, shall we?

Currently:

Player 1 has : £2000

Player 2 has : £2000

Player 3 has : £2000

Player 4 has : £2000

In their pots.

Question 1: [Q1] What is the chemical symbol for water? [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 2's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 3's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 4's go:

What is your attempt number: 1?

H2O

That is the right answer.

Currently:

Player 1 has : £4000

Player 2 has : £4000

Player 3 has : £4000

Player 4 has : £4000

In their pots.

Question 2: [Q2] Fill in the blank: The _____ Ocean is the largest ocean on earth. [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 2's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 3's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 4's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Currently:

Player 1 has : £8000

Player 2 has : £8000

Player 3 has : £8000

Player 4 has : £8000

In their pots.

Question 3: [Q3] What country gifted the famous 'Statue of Liberty' to the USA?

[ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 2's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 3's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 4's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Currently:

Player 1 has : £16000

Player 2 has : £16000

Player 3 has : £16000

Player 4 has : £16000

In their pots.

Question 4: [Q4] What is the only mammal capable of true flight? [ALL CAPITALS]

| [PLURAL ANSWER WANTED]

Player 1's go:

What is your attempt number: 1?

BATS

That is the right answer.
Player 2's go:
What is your attempt number: 1?

BATS

That is the right answer.
Player 3's go:
What is your attempt number: 1?

BATS

That is the right answer.
Player 4's go:
What is your attempt number: 1?

BATS

That is the right answer.
Player 1 has earned: £32000
Player 2 has earned: £32000
Player 3 has earned: £32000
Player 4 has earned: £32000
Well done to all!

3.1.2 Test Plan: Test 2

If the following input is given by the user when asked in turn:

New | NEW | | Player 1: H2O, PACIFIC, FRANCE, BAT | Player 2: Dunno [x2], PACIFIC, |
Player 3: Dunno [x2], Dunno [x2] | Player 4: Dunno [x2], Dunno [x2] SUDDEN STOP ...

```
[5]: quiz6.main(null);
```

Type NEW if you want to start a new quiz or LOAD if you already have a saved file.

New

ONLY INPUT 'NEW' OR 'LOAD'. AND MAKE SURE IT'S IN CAPITALS

Type NEW if you want to start a new quiz or LOAD if you already have a saved file.

NEW

Currently:

Player 1 has : £2000
Player 2 has : £2000
Player 3 has : £2000
Player 4 has : £2000

In their pots.

Question 1: [Q1] What is the chemical symbol for water? [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

H2O

That is the right answer.

Player 2's go:

What is your attempt number: 1?

Dunno

That is the wrong answer.

What is your attempt number: 2?

Dunno

That is the wrong answer.

Player 3's go:

What is your attempt number: 1?

Dunno

That is the wrong answer.

What is your attempt number: 2?

Dunno

That is the wrong answer.

Player 4's go:

What is your attempt number: 1?

Dunno

That is the wrong answer.

What is your attempt number: 2?

Dunno

That is the wrong answer.

Currently:

Player 1 has : £4000

Player 2 has : £500

Player 3 has : £500

Player 4 has : £500

In their pots.

Question 2: [Q2] Fill in the blank: The _____ Ocean is the largest ocean on earth. [ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.

Player 2's go:

What is your attempt number: 1?

PACIFIC

That is the right answer.
Player 3's go:
What is your attempt number: 1?

Dunno

That is the wrong answer.
What is your attempt number: 2?

Dunno

That is the wrong answer.
Player 4's go:
What is your attempt number: 1?

Dunno

That is the wrong answer.
What is your attempt number: 2?

Dunno

That is the wrong answer.
Currently:
Player 1 has : £8000
Player 2 has : £1000
Player 3 has : £125
Player 4 has : £125
In their pots.
Question 3: [Q3] What country gifted the famous 'Statue of Liberty' to the USA?
[ALL CAPITALS]
Player 1's go:
What is your attempt number: 1?

```
| quiz6.main(null);  
Evaluation interrupted.
```

3.1.3 Test Plan: Test 3

If the following input is given by the user when asked in turn:

LOAD | | Player 1: FRANCE, BATS | Player 2: FRANCE, BATS | Player 3: FRANCE, BATS |
Player 4: Dunno [x2], BATS ...

```
[6]: quiz6.main(null);
```

Type NEW if you want to start a new quiz or LOAD if you already have a saved file.

LOAD

Currently:
Player 1 has : £8000

Player 2 has : £1000

Player 3 has : £125

Player 4 has : £125

In their pots.

Question 3: [Q3] What country gifted the famous 'Statue of Liberty' to the USA?
[ALL CAPITALS]

Player 1's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 2's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 3's go:

What is your attempt number: 1?

FRANCE

That is the right answer.

Player 4's go:

What is your attempt number: 1?

Dunno

That is the wrong answer.

What is your attempt number: 2?

Dunno

That is the wrong answer.

Currently:

Player 1 has : £16000

Player 2 has : £2000

Player 3 has : £250

Player 4 has : £31

In their pots.

Question 4: [Q4] What is the only mammal capable of true flight? [ALL CAPITALS]
| [PLURAL ANSWER WANTED]

Player 1's go:

What is your attempt number: 1?

BATS

That is the right answer.

Player 2's go:

What is your attempt number: 1?

BATS

That is the right answer.
Player 3's go:
What is your attempt number: 1?

BATS

That is the right answer.
Player 4's go:
What is your attempt number: 1?

BATS

That is the right answer.
Player 1 has earned: £32000
Player 2 has earned: £4000
Player 3 has earned: £500
Player 4 has earned: £62
Well done to all!

END OF LITERATE DOCUMENT